



Critical success factors of users' continuous intention of adopting cryptocurrency exchanges: LAS-VICT principle

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Abstract

The proliferation of cryptocurrencies has contributed to the emergence of different cryptocurrency exchanges (crypto-exchanges). While these services may be regarded as FinTech, involving cryptocurrency as the major transaction currency has made these services potentially distinctive from other fiat-based FinTech services. Thus, the critical success factors of crypto-exchanges may not be identical to those of other fiat-based FinTech services. Grounded on theories related to FinTech and service varieties, we developed a survey and explored the role of different factors on users' continuous intention of adopting the crypto-exchanges. Our results suggested that when users perceive specific crypto-exchange characteristics, they are more likely to adopt the exchange continuously. Combining previous literature, we name this set of factors "LAS-VICT principle", including low user-burden, asset-light, scalability, variety, innovativeness, scalability, and transparency. However, users' emphasis on factors may differ based on their cryptocurrency experience. Based on our findings, we provided some theoretical and practical implications.

Keywords Bitcoin · Cryptocurrencies · Cryptocurrency exchanges · FinTech · Binance

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Introduction

Cryptocurrencies (or simply "crypto") have gained increasing popularity in recent years, given they support cheaper, faster (Hendershott et al., 2021), and geographically independent (Abramova & Böhme, 2016) transactions. As of June 2023, there were approximately 516 million identity-verified cryptocurrency asset owners (Statista, 2023). Cryptocurrencies have been well-known as emerging investment options, but they also have huge potential in fostering financial inclusion and are increasingly accepted for daily-life payment for products and services (Hsu & Au, 2023). For example, cryptocurrencies were used by Binance, a leading cryptocurrency exchange (crypto-exchange), to support Ukrainian refugees during the Russian invasion via their cryptocurrency-based debit cards (Binance, 2023). Researchers have predicted more pragmatic and diversified cryptocurrency applications in the future and have called for more research in this area (Li & Whinston, 2020).

As issues such as key management and the lack of user-friendly interfaces (Fröhlich et al., 2021) may become adoption obstacles, the emergence of cryptocurrencies-related

businesses, such as exchanges and wallet services, has facilitated users' adoption (Dierksmeier & Seele, 2018; Kazan et al., 2015). Some highly successful crypto-exchanges, such as Binance and Coinbase, have integrated the functions of exchanges and cryptocurrency wallets, hoping to address users' needs more comprehensively. Some of them have even launched cryptocurrency debit cards, which allow cryptocurrencies to be spent as if using conventional credit cards (Au et al., 2022; Shieh et al., 2023). Moreover, beyond investment and speculation, cryptocurrencies are increasingly adopted and integrated into the products and services of different firms, such as Alternative Airlines and Shopify. According to Cryptwek.com, Bitcoins are accepted by more than 9,000 merchants, while Ethereum, being the second most popular cryptocurrency in the world, are accepted by more than 5,000 merchants. In both cases, the cryptocurrency-accepting merchants are offering different types of products and services, such as Internet services, fashions, tourism, gaming, luxury, and gambling, while the number of these merchants has constantly been increasing (Cryptwerk-Pay with Bitcoin, 2024).

Existing literature on cryptocurrencies largely focused on the technical, macroeconomic, or individual adoption aspects (see, e.g., Hai et al., 2023). And yet, despite the emergence of crypto-exchange businesses, previous research on cryptocurrencies seems less considering the business successes of these services. While cryptocurrencies may be regarded as Financial Technologies (FinTech), considerable differences exist between mainstream FinTech and cryptocurrency applications. For example, social and ethical concerns over cryptocurrency applications are more common than similar concerns over other FinTech applications (Dierksmeier & Seele, 2018), affecting potential users' perception and adoption (Borri & Shakhnov, 2020). Other fundamental differences between crypto-applications and other FinTech applications include the involved currencies (i.e., cryptocurrencies vs. fiat currencies, with the former being more volatile in price and yet to be widely adopted) and the associated policies and regulations. Price volatility that is normal for crypto-exchanges and their users may be considered disastrous in non-crypto financial market contexts. Moreover, most, if not all, FinTech applications operate mainly in a single market, but many cryptocurrency applications are concurrently serving different countries (e.g., Binance and Coinbase, see Xia et al., 2020). Finally, yet importantly, considering recent significant changes in the FinTech landscape (Fu & Mishra, 2022) and the difference between crypto-exchanges and non-crypto FinTech businesses (e.g., the difference in transaction currencies among these businesses), existing theories related to FinTech business successes may not adequately be applied in crypto-exchange contexts. A lack of attention to this research gap may lead to the inaccurate formulation of business strategies

by crypto-exchanges. Cryptocurrencies' benefits to other needy people may also be compromised.

Focusing on crypto-exchanges instead of cryptocurrencies, we designed this study based on different FinTech literature, particularly the LASIC Principles proposed by Lee and Teo (2015), which suggested that the five critical success factors (CSFs) of successful FinTech firms (low margin, asset-light, scalable, innovative, and compliance easy), and has been cited by numerous FinTech literature (e.g., Burtch et al., 2018; Campanella et al., 2023). Our study may guide crypto-exchanges in developing their businesses. Accordingly, our research question is, "*What are the critical success factors (CSFs) of customers' continuous intention of crypto-exchange adoptions?*".

We test the impact of different CSFs related to FinTech on users' continuous adoption of crypto-exchanges, and arrive at the LAS-VICT principles (low user-burden, asset-light, scalability, Variety, innovativeness, scalability, and transparency), an expansion of the LASIC principle. Based on our findings, we provided a range of theoretical implications in FinTech and practical implications for crypto-exchanges.

Literature review on cryptocurrencies and crypto-exchanges

Cryptocurrency is a type of digital money that relies on cryptography to secure token transfers and record transactions on a decentralised digital register. Its roots may be traced back to the blockchain technology introduced by Satoshi Nakamoto, which enabled Bitcoin's launch in 2009 (Li & Whinston, 2020). In recent years, cryptocurrencies have been adopted not only among a niche group of technology pioneers but also in e-commerce contexts (Dyosi & Tanner, 2023). Previous studies on cryptocurrencies can be categorised into three streams, including (1) enabling factors of adoption, (2) barriers to adoption, and (3) business opportunities (see Table 1).

Previous literature indicated that cryptocurrency applications might enhance financial inclusion and reduce transaction time and costs. The need to overcome knowledge complexity has driven the emergence of some crypto-exchanges, with some achieving global successes and being adopted as venues for studying cryptocurrencies' price trends and volatility (e.g., Ji et al., 2021).

In this study, we focused on crypto-exchanges, which we define as trading platforms for acquiring or selling cryptocurrencies for other assets and, at the same time, providing other online financial services, with examples like Binance, Uphold, and Coinbase. While some financial trading and exchanges (e.g., TradeRepublic) have recently stepped into the landscape of cryptocurrencies, our study focused on exchanges with cryptocurrency-related products/services

Table 1 A selected list of arguments around cryptocurrencies

Dimensions	Descriptions
Enabling factors of cryptocurrencies applications	
Financial inclusion	Cryptocurrencies may help reduce daily financial transaction costs and hyperinflations (Dierksmeier & Seele, 2018) and thus, harness financial inclusion, foster human development, and address poverty, especially in politically unstable countries (Lee & Teo, 2015; Qureshi & Xiong, 2018)
Lowered transaction fees	The development of the cryptocurrency market offers a significant opportunity to reduce transaction fees (Mendoza-Tello et al., 2019; Zohar, 2015), particularly in the case of cross-border transactions (Carvalho et al., 2020)
Reduced transaction time	With blockchain technologies behind, cryptocurrencies may reduce data transmission and shorten the time required for different steps of financial transactions (Hendershott et al., 2021)
Barriers to the adoption of cryptocurrencies	
Social and legal concerns	Cryptocurrencies have been accused of financing illegal activities, such as scams, fake crypto-exchanges (Xia et al., 2020), money laundering (Brenig et al., 2015), and tax evasion (Dierksmeier & Seele, 2018). Also, given the advantages and opportunities brought by cryptocurrencies, more people are mining cryptocurrencies, which create significant electricity consumption (Li et al., 2019). These issues have become a concern of different regulatory bodies (Chason, 2019), which may restrict wider cryptocurrency adoptions (Borri & Shakhnov, 2020)
Price volatility	With no central issuing authorities, many cryptocurrencies may have value primarily based on market forces and may thus fluctuate significantly (Ciaian et al., 2016; Qureshi & Xiong, 2018). Occasionally, transaction fees also fluctuate heavily (Ilk et al., 2021)
Adoption difficulties	The difficulties of key management, the lack of user-friendly interfaces, and cryptocurrencies-related knowledge may resist cryptocurrency adoption (Fröhlich et al., 2021). Addressing these challenges is essential for ensuring users' continuous service adoption (Chang et al., 2016)
Diversified viewpoints on cryptocurrency valuation	Various economic schools of thought and potential users perceive the value of cryptocurrencies differently. Some users may highly regard cryptocurrencies, but others may treat cryptocurrencies as valueless (Grassman et al., 2021; Treiblmaier, 2022)
Business opportunities	
Cryptocurrencies wallet products and services	Given the popularity of cryptocurrencies, cryptocurrency wallet (crypto-wallet) services are increasingly important for cryptocurrency management, which helps organise and store private keys remotely while ensuring cryptocurrencies' rapid accessibility (Biryukov & Tikhomirov, 2019). Some of these wallets may be integrated with crypto-exchanges or other services to address users' needs more comprehensively (Kazan et al., 2015)
Enabling new and impacting business models	The emergence of cryptocurrencies has empowered new business models and impacted existing ones (Dierksmeier & Seele, 2018). In addition to crypto-wallet services (Li & Whinston, 2020), cryptocurrencies have enabled variants of existing FinTech business models (Weking et al., 2020)

as their primary business focus, not non-crypto trading and exchange platforms that tenuously stepped into the landscape of cryptocurrencies. However, these crypto-exchanges are not only platforms for trading cryptocurrencies but also provide a spectrum of FinTech functions, such as mobile payment and money transfer. As a result, we consider crypto-exchanges to be non-traditional FinTech products.

Despite the increasingly competitive landscape, the development and successes of crypto-exchanges are less commonly addressed. Previous articles on cryptocurrencies published in prominent information systems (IS) journals were more related to the technical side (Lagna & Ravishankar, 2022). While we may treat crypto-exchanges as a type of FinTech, cryptocurrency applications are unique

from other FinTech applications in many aspects (Hamm, 2023). For example, there are price volatility and electricity wastage concerns around cryptocurrencies. In turn, price volatility lowers market efficiency and liquidity (Wei, 2018) and creates speculative opportunities for investors, which are less common in non-cryptocurrencies investments (Mendoza-Tello et al., 2019). Comparing the crypto-exchanges and other financial markets or exchanges, it is not only the primary currencies involved that are different (i.e., fiat vs. crypto). However, crypto-exchanges are more likely to launch innovative service options to engage more customers (Shieh et al., 2023), and are involved in scams (Xia et al., 2020). These differences have not only made cryptocurrencies yet to be widely adopted (Abooleet & Fang, 2023), but

Table 2 A selected list of attributes related to FinTech business successes

Attributes	Description
Low user-burden	Profit margins should be kept low to establish a critical mass of users and, thus, a high network effect. In turn, profitability may be attained by high demand for the core services and other channels, such as advertising (Lee & Teo, 2015). Studying in different contexts, Singh (2020) and Arslanian and Fischer (2019) argued low margins are a critical part of many FinTech firms. However, in addition to offering low-cost services, FinTech firms should offer benefits and convenience. Collectively, low user burden is a key attribute for successful FinTech firms (Nurbaev et al., 2022)
Asset-light	By riding on existing infrastructure, asset-light FinTech businesses can be innovative and scalable without incurring significant fixed costs on assets. In turn, users can enjoy financial services at a low cost (Lee & Teo, 2015). In addition, traditional financial institutions usually have a lower profit-asset ratio than other FinTech, such as neobanks and crypto-exchanges (Nurbaev et al., 2022)
Scalable	FinTech businesses and their core technologies may start small but need to be scalable without drastically increasing costs or compromising efficiency to reap the benefits of network externalities. For example, this may be done by getting businesses done online instead of through physical outlets (Lee & Teo, 2015). Muthukannan et al. (2021) highlighted the importance of scalability in FinTech ecosystems and recommended some measures, while Bruton et al. (2015) highlighted the adoption of platform-mediated approaches by FinTech firms for handling the huge transaction volume
Variety	Variety, which refers to the number of product or service options in the same company, can increase customers' likelihood of patronising a company (Kahn, 1995). The provision of products/services variety can be part of a prosperous FinTech ecosystem, which may fulfil clients' needs more comprehensively (Leong et al., 2017). In addition, crypto-exchanges product and service options may be classified into three categories: deposit, wealth growth, and withdrawal (Shieh et al., 2023)
Innovativeness	Successful FinTech businesses should innovate their products and operations (Gimpel et al., 2018; Liu et al., 2020). For example, popularising smartphones and their related technologies (e.g., contactless technologies) has enabled more FinTech innovations (Lee & Teo, 2015). On the other side, users' innovativeness may also drive their FinTech adoption (Setiawan et al., 2021)
Compliance easy	Businesses not subject to high compliance regimes may focus more on innovations, as they may face lower capital requirements and spend fewer resources on compliance activities (Lee & Teo, 2015)
Transparency	Transparency impacts users' propensity to switch to FinTech options (Jünger & Mietzner, 2020). By enhancing transparency, FinTech firms could mitigate the risk of financial instability (Fung et al., 2020)

also indicate that CSFs of crypto-exchanges may be potentially different from other FinTech applications. Therefore, a refreshed understanding of crypto-exchange business successes is needed so that these exchanges, and thus the future of cryptocurrencies, may be sustained.

Theoretical foundation

FinTech can be understood as a broad umbrella term for disruptive technologies in the financial sector, which may include third-party payment, insurance products, risk management, and peer-to-peer (P2P) lending (Shim & Shin, 2016). While both FinTech and cryptocurrencies have become buzzwords in recent years, FinTech utilises technology for financial services, while cryptocurrency is a digital asset that often drives FinTech innovations (such as crypto-exchanges) and enables decentralised transactions. However, not all FinTech involves cryptocurrencies. Researchers are interested in studying the factors behind the success of FinTech. Among different FinTech literature, LASIC Principles proposed by Lee and Teo (2015) is one of the most widely cited frameworks, which suggested five attributes of FinTech business success: low margin, asset-light, scalable, innovative, and compliance

easy. Based on this framework, some other researchers have expanded and modified the framework to enhance the understanding of FinTech business success (see Table 2 for more details about the attributes more commonly highlighted by previous literature).

Yet, previous literature has revealed some limitations. For example, Lee and Teo (2015) established their arguments based on only two m-payment FinTech firms, which implied that their arguments might be subject to further validation in other FinTech contexts, such as crypto-exchanges, especially given the fundamental differences between crypto-exchanges and other FinTech applications, such as the subject transaction currencies (i.e., fiat currencies vs cryptocurrencies) and wider ethical concerns for cryptocurrencies-related businesses (Shieh et al., 2023). In addition, while some literature argues variety as an attribute of successful FinTech businesses, we may also observe some successful FinTech businesses with monopolistic service offerings (Au et al., 2020). In other words, there remain some inconsistencies regarding the role of a variety of service offerings in FinTech businesses. Afterall, given the significant changes and development in the FinTech landscape (Fu & Mishra, 2022), we acknowledge the contribution of previous literature, but revising or testing of existing arguments in another FinTech context (i.e., crypto-exchanges) will be desirable.

With users' perception of these CSFs as independent variables (IVs), we adopted task-technology fit (TTF)¹ and Perceived Usefulness (PU)² as dependent variables (DVs), both commonly used for technology adoption. More specifically, we argue that users may perceive the crypto-exchanges with higher TTF by realising these characteristics. According to the TTF theory, an information system (IS) is more likely to positively impact individuals' performance and be used if the capabilities of the IS match users' tasks. In addition, TTF is closely linked with technology characteristics (Goodhue & Thompson, 1995), which has been adopted in previous literature on e-commerce (e.g., Dishaw & Strong, 1999; Gebauer et al., 2010) and e-learning (e.g., Isaac et al., 2019; Khan et al., 2018). Also, previous literature suggested different factors that may impact PU, a crucial construct of the technology acceptance model (Davis, 1989; Davis et al., 1989), including information transparency (Al-Jabri & Roztocki, 2015) and personal innovativeness (Fagan et al., 2012). More specifically, in FinTech contexts, previous literature has also highlighted the role of information transparency (see e.g., Fung et al., 2020) and personal innovativeness (e.g., Setiawan et al., 2021).

We aim to statistically validate the potential characteristics of a successful crypto-exchange from user perspectives. For the attributes of low user burden, we tested the cost and benefits of using the crypto-exchanges separately. As indicated in Table 2, profit margins (Arslanian & Fischer, 2019; Singh, 2020) are essential to lowering user burdens (Nurbaev et al., 2022). Therefore, having a high financial cost (such as a high transaction cost) and providing financial incentives (such as coupons) for using crypto-exchanges would reduce and increase their profit margins and will affect the users' perceived usefulness and task-technology fit (See-To & Ho, 2016). While some previous IS literature has discussed the negative influence of costs on perceived usefulness, the usage of crypto-exchanges generates both financial costs and benefits, which is different from other IS. Also, in the era of platformisation (Au, 2020), many IS may be used free of charge. Given many crypto-exchanges operate in forms of digital platforms, cost may not always be relevant to crypto-exchanges' adoption. Therefore, we test the role of both perceived users' financial benefits and costs on perceived usefulness and task-technology fit in the context of crypto-exchanges, and proposed,

H1a: Perceived users' financial costs negatively influence perceived usefulness.

H1b: Perceived users' financial costs negatively influence task-technology fit.

H2a: Perceived users' financial benefits positively influence perceived usefulness.

H2b: Perceived users' financial benefits positively influence task-technology fit.

In addition, grounded on previous literature about the role of variety in users' technology and service adoption (Kahn, 1995; Leong et al., 2017), we tested the role of varieties on TTF and PU. We split variety into three aspects (deposit, wealth growth, and withdrawal) based on the study of Shieh et al. (2023) on 14 crypto-exchanges. Shieh et al. (2023) also mentioned that the variety should include some options that are friendly for new users so that they are less likely to be resisted by technology complexity while the needs of different users can be more comprehensively covered. Altogether, we proposed,

H3a: Perceived variety of deposit options positively influences perceived usefulness.

H3b: Perceived variety of deposit options positively influences task-technology fit.

H4a: Perceived variety of wealth growth options positively influences perceived usefulness.

H4b: Perceived variety of wealth growth options positively influences task-technology fit.

H5a: Perceived variety of withdrawal options positively influences perceived usefulness.

H5b: Perceived variety of withdrawal options positively influences task-technology fit.

On the other hand, scalability is another characteristic of successful FinTech firms, which refers to capabilities to increase in scale without drastically increasing costs or compromising the efficiency of the technology (Lee & Teo, 2015), and is related mainly to the interfaces of serving customers (Bruton et al., 2015). In particular, as shown by Lee and Teo (2015), we treat perceived scalability as whether crypto-exchanges can provide service to accommodate more customers without reducing service quality. We assume that long-term end users of crypto-exchanges may, to some extent, witness how the crypto-exchanges scale up and develop by, for example, observing the growth of the number of service offerings, considering statistics related to the crypto-exchanges and reading 3rd-party news reports related to the crypto-exchanges. Therefore, the scalability of a crypto-exchange may be perceived by its users. Accordingly, we propose,

¹ Task-technology fit (TTF) assesses how well a technology matches and supports specific tasks or activities within an organization, enhancing efficiency and effectiveness (Goodhue and Thompson 1995).

² Perceived usefulness (PU) refers to the degree to which consumers believe that a particular technology will facilitate the transaction process (Davis 1989).

H6a: Perceived scalability positively influences perceived usefulness.

H6b: Perceived scalability positively influences task-technology fit.

Also, innovativeness, which is related to using newer technologies for products and operations to deliver the value to customers, has been emphasised as part of the LASIC principle (Lee & Teo, 2015) and in other FinTech literature (Gimpel et al., 2018; Liu et al., 2020). Some literature, specifically in crypto-exchange contexts (e.g., Au et al., 2022; Shieh et al., 2023), has suggested the need for innovativeness for exchanges to develop different service options to address users' needs. In this study, we followed Hwang et al. (2019), which suggested firms' innovativeness can be evaluated by their customers, in measuring the perceived innovativeness of crypto-exchanges through their service uniqueness, newness, and creativity.

H7a: Perceived innovativeness positively influences perceived usefulness.

H7b: Perceived innovativeness positively influences task-technology fit.

Furthermore, we argue that crypto-exchanges will not only follow the legislation regarding the compliance aspects. Conversely, scholars and the general public have been concerned about its social impact, such as electricity wastage (Li et al., 2019). In broader FinTech contexts, the influence of legal risk in FinTech adoption is also highlighted by some previous literature (e.g., Ali et al., 2021). Moreover, we should also note the emergence of ethical consumption, which fosters customers' increasing emphasis on whether the firms comply with the legal requirements and social expectations before patronising them (Bray et al., 2011). Crypto-exchanges, accordingly, will respond to these legal and social concerns in their business (Au et al., 2022). Therefore, we expand the compliance consideration into two aspects, including (1) social compliance and (2) legal compliance. Altogether, we proposed,

H8a: Perceived legal compliance positively influences perceived usefulness.

H8b: Perceived legal compliance positively influences task-technology fit.

H9a: Perceived social compliance positively influences perceived usefulness.

H9b: Perceived social compliance positively influences task-technology fit.

Transparency is another CSF commonly emphasised in different IS contexts, such as enterprise resources planning systems (Al-Jabri & Roztock, 2015). In broader FinTech

contexts, such as neobanks (Nurbaev et al., 2022), transparency can impact users' propensity to switch to FinTech options (Jünger & Mietzner, 2020), and thus contribute to the overall success of FinTech firms (Werth et al., 2023). Moreover, in cryptocurrency contexts, Miraz et al. (2022) suggested that transparent transactions drive intention to use, which is highly related to TTF and PUSE. Therefore, we proposed,

H10a: Perceived transparency positively influences perceived usefulness.

H10b: Perceived transparency positively influences task-technology fit.

Beyond the proposed variables, previous literature suggested that PU may contribute to users' intention to use the technologies (Lu et al., 2003; Yoon, 2009), while TTF may contribute to CI (Smith et al., 2015). We emphasise the continuance of the intention since the long-term success of crypto-exchanges will seek customers' continuous usage. In FinTech contexts, a single transaction can only bring limited revenue to the firm. Conversely, when customers continuously use the services, a higher amount of revenue can be gradually accumulated, and thus, the FinTech firm is more likely to succeed. Therefore, we proposed,

H11: Perceived usefulness positively influences continuance intention of adoption.

H12: Task-technology fit positively influences continuance intention of adoption.

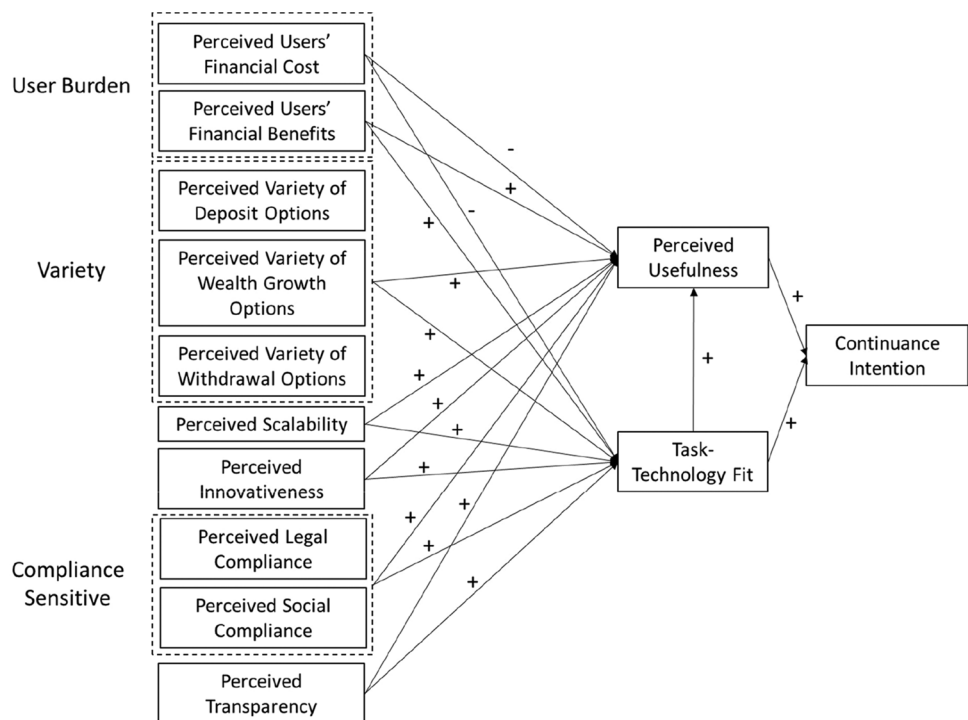
Moreover, Dishaw and Strong (1999) integrated the concept of TTF into the TAM model and suggested the role of task-technology fit on perceived usefulness. In FinTech contexts, Sharif and Naghavi (2021) also argued that in online financial trading contexts, TTF can contribute to perceived usefulness. Testing the hypothesis in crypto-exchange contexts, we proposed,

H13: Task-technology fit positively influences perceived usefulness.

Altogether, we have developed the following theoretical model (see Fig. 1).

Methodology

We used Amazon Mechanical Turk (MTurk) to collect our data via surveys, which allows us to access hard-to-reach subjects (Smith et al., 2015) while avoiding oversampling participants from WEIRD (i.e., Western, educated, industrialised, rich, and democratic) backgrounds (Jia et al., 2017).

Fig. 1 Proposed theoretical model

It is considered comparatively effective as traditional data collection approaches (Maier et al., 2019; Mattke et al., 2021) and is increasingly accepted among senior basket IS journals (e.g., Mattke et al., 2021). We also considered the potential issues of adopting MTurk for data collection, including self-selection bias, attentiveness, ability, and social desirability (Cheung et al., 2017; Jia et al., 2017). Furthermore, to ensure respondents were real crypto-exchange users, we first asked them to specify a crypto-exchange they were using, followed by asking them to upload a screen capture of their accounts on the crypto-exchange (after masking all personally identifiable information). We have also asked them to tick that they are not affiliated with the selected crypto-exchanges (other than being a user). In turn, we can filter ineligible participants and ensure their sincere, accurate, and reliable answers without being affected by payment incentives (Antin & Shaw, 2012). For example, respondents would be rejected if they selected a FinTech firm not primarily targeted at the landscape of cryptocurrencies (e.g., Tinkoff, eToro) or if they uploaded a fake screenshot (e.g., which was downloaded from a third-party website or showed a price of cryptocurrency derivated largely from the market price by the time they accomplished the survey). We have also included some attention confirmation questions to ensure they paid adequate attention when completing our survey. Our pilot test by IS postgraduates indicated that it took approximately 10 min to finish this survey, so we believe there should be no fatigue issues.

We asked respondents to specify their most frequently used crypto-exchanges at the beginning of the survey and

answer the questions based on this crypto-exchange. As a preamble, we explicitly mentioned that our survey focused on crypto-exchange services but not cryptocurrency investment at the beginning to ensure that our participants knew that they were reviewing crypto-exchanges instead of cryptocurrencies. These measurement items were developed based on previous literature, with modifications to fit our research context. Except for control variables, measurement items are on a 9-point Likert scale (see Appendix A for measurement item details, including the citation of previous literature we adopted to establish our measurement items).

The data was collected in February 2023. This was three months after the collapse of FTX³ and nine months after the collapse of Terra-LUNA.⁴ We are aware these events have negatively influenced the public interest in cryptocurrencies and thus, the adoption of crypto-exchanges. We believe that, at our data collection time, the demotivating influence of these incidents was moderated. Respondents were relatively

³ FTX is a company that operated a cryptocurrency exchange. At its peak in July 2021, had over one million users and was the third-largest cryptocurrency exchange by volume (Reuters, 2021). However, in November 2022 FTX has been in bankruptcy proceedings in the USA (Yaffe-Bellany, 2022).

⁴ Terra-LUNA refers to a cryptocurrency ecosystem. Terra is a blockchain protocol and payment platform used for algorithmic stablecoins. The project was created in 2018 by Terraform Labs and is most known for its Terra stablecoin (UST) and the associated Luna reserve asset cryptocurrency (LUNA). In May 2022, the value of both UST and LUNA crashed to almost zero, wiping out almost \$45 billion of market capitalisation (Yaffe-Bellany & Griffith, 2022).

calmer, with their answers less affected by unfavorable incidents. Also, we are focusing on the functions and service offerings of the centralised crypto-exchanges, which will not be changed by the price volatility of cryptocurrency alone.

Among all respondents, 213 provided valid samples, with 59 female and 152 male respondents (2 others did not indicate their gender types). The average age was 33.7 (with a standard deviation of 9.41). The average number of months of experience in cryptocurrency investment/trading was 22.7 (with a standard deviation of 21.4), while the average self-reported cryptocurrency trading frequency (on a 9-point scale, with 1 meaning “very rarely” and 9 meaning “very frequently”) was 5.34 (with a standard deviation of 2.35). Furthermore, around 53% of our participants earned an associate or bachelor degree, with nearly an equal split of high school graduates and people with graduate degrees for the rest of the participants. Considering the number of variables in the theoretical model and the recommendation of Hair et al. (2010), we collected adequate samples to validate our model.

For users’ geographical locations, 112 respondents came from the USA, 37 from India, 20 from Italy, 15 from Brazil, 6 from the United Kingdom, 3 from Canada, 2 from France, and 2 from Mexico. In addition, 8 were reported from other countries (only one respondent from each of these countries), and 8 did not report their geographical locations. The diversified nationality backgrounds of our respondents echoed our choice of using MTurk, which would help to improve the quality of data collected, i.e., not focusing on WEIRD.

Moreover, the crypto-exchanges that more than one respondent selected included Coinbase (77), Binance (57), Crypto.com (19), Uphold (5), Robinhood (5), Wazirx (5), Metamask (3), KuCoin (3), Giottus (2), Cashapp (2), BlockFi (2), Gemini (2), and X (2). Only one respondent selected another 28 crypto-exchanges. The high popularity of Coinbase, Binance, and Crypto.com has been reflected in our sample, while the choice difference among respondents ensured results were not skewed towards any particular crypto-exchanges. Overall demographic information indicated that we included a diversified range of respondents. In addition, Appendix B shows the descriptive statistics of each construct and the correlation matrix between these constructs, while Appendix C shows the factor loadings of each construct.

Results

We used Harman’s One-Factor Test to verify if our dataset suffered from common method bias. The total variance of the single factor was 35.47%. Lower than the 50% threshold, suggesting common method bias is not a problem in our case

(Podsakoff et al., 2012). After removing INN2, which has a low loading to the corresponding construct, the results confirmed both convergent validities as all t -values of the factor loadings are significant with $p < 0.01$ and all the Cronbach’s α values > 0.70 . The discriminant validity is also achieved as all instrument items have a loading > 0.70 on their associated factors and low loading on other factors (Nunnally, 1978).

Given that experiences may differ in technological tool usage and cryptocurrencies are still emerging, we split respondents into two groups by the median of the product of self-reported frequency and months of experience (i.e., 72). Prior research on online purchases (Wu et al., 2022) and investment (Holm & Rikhardsson, 2008) showed that experienced and inexperienced users would have different behaviour. Therefore, we coded our respondents with a value higher than or equal to 72 as experienced users, while the rest were inexperienced users in our data analysis to examine this effect. We used SmartPLS 4.0 (Ringle et al., 2022) to validate our theoretical model based on the two data subsets and the overall dataset. In the optimised model (see Table 3), we removed those insignificant paths from the model by removing the most insignificant path of the overall users’ dataset (starting from the highest p -value) one by one. A construct with a high p -value was kept if the path was significant for one of the sub-groups (inexperienced or experienced users).

As per the indicated R -square values (all $> 50.0\%$), our optimised model’s IVs (characteristics under-test) may collectively explain the variance of perceived usefulness (PU) and task-technology fit (TTF) to a reasonable extent. The impact of perceived financial costs and taxation burden (FIN-C) and perceived legal compliance (COMP-L) did not influence our model and were taken from our optimised model. Thus, H1a and H1b related to FIN-C and H8a and H8b related to COMP-L were rejected. On the contrary, perceived financial benefit (FIN-B) and transparency (TRA) positively influenced the overall sample of PU and TTF. Therefore, H2a and H2b related to FIN-B and H10a and H10b related to TRA were supported. In addition, TRA positively influenced inexperienced and experienced users in the TTF (H10b) and inexperienced users in PU (H10a). In addition, the perceived variety of withdrawal options (PVW) positively influenced TTF (H5b) for the overall sample and marginally positively influenced PU ($p = 0.051$) (H5a) for inexperienced users.

We also noted that two constructs only influenced TTF but not PU, i.e., perceived scalability (SCAL, for overall and experienced users, H6b) and perceived variety of growth options (PVG, for overall and inexperienced users, H4b). On the other hand, three constructs only influenced PU, i.e., perceived variety of deposit options (PVD, marginally negative for inexperienced users, $p = 0.099$, H3a), perceived innovativeness (INN, negative for inexperienced users, and

Table 3 Results of the optimised model

DV	IV	Overall users		Inexperienced users		Experienced users	
		Path coeff	<i>p</i> -value	Path coeff	<i>p</i> -value	Path coeff	<i>p</i> -value
	CI	Adjusted <i>R</i> -square: 68.7%		Adjusted <i>R</i> -square: 62.0%		Adjusted <i>R</i> -square: 75.9%	
H11	PU	0.164	0.088 #	0.161	0.212	0.206	0.074 #
H12	TTF	0.698	<0.001 ***	0.668	<0.001 ***	0.700	<0.001 ***
	TTF	Adjusted <i>R</i> -square: 67.0%		Adjusted <i>R</i> -square: 70.8%		Adjusted <i>R</i> -square: 52.8%	
H2b	FIN-B	0.120	0.01 **	0.091	0.138	0.124	0.152
H4b	PVG	0.185	0.014 *	0.260	0.001 **	0.135	0.269
H5b	PVW	0.129	0.03 *	0.161	0.113	0.091	0.198
H6b	SCAL	0.156	0.018 *	0.111	0.227	0.215	0.030 *
H10b	TRA	0.456	<0.001 ***	0.424	<0.001 ***	0.445	0.002 **
	PU	Adjusted <i>R</i> -square: 65.0%		Adjusted <i>R</i> -square: 61.5%		Adjusted <i>R</i> -square: 67.0%	
H2a	FIN-B	0.123	0.044 *	0.081	0.278	0.106	0.157
H3a	PVD	−0.103	0.346	−0.266	0.099 #	0.057	0.628
H5a	PVW	0.090	0.367	0.322	0.051 #	−0.122	0.188
H7a	INN	−0.108	0.069 #	−0.184	0.019 *	0.032	0.718
H9a	COMP-S	0.229	0.005 **	0.308	0.002 **	0.157	0.264
H10a	TRA	0.199	0.004 **	0.278	0.008 **	0.124	0.130
H13	TTF	0.510	<0.001 ***	0.381	0.003 **	0.582	<0.001 ***

1. DV, dependent variable; IV, independent variable; CI, continuance intention; TTF, task-technology fit; PU, perceived usefulness; FIN-C, perceived users' financial cost; FIN-B, perceived users' financial benefits; SCAL, perceived scalability; PVD, perceived variety of deposit options; PVG, perceived variety of wealth growth options; PVW, perceived variety of withdrawal options; INN, perceived innovativeness; COMP-L, perceived legal compliance level; COMP-S, perceived social compliance level; TRA, perceived transparency

2. The following constructs did not influence the model: COMP-L and FIN-C

3. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$, # $p < 0.10$

marginally negative for overall users, $p = 0.069$, H7a), and perceived social compliance (COMP-S, positive for overall and inexperienced users, H9a). In addition, we observed the positive impact of TTF on PU for all situations, and so H13 was supported.

Finally, yet importantly, both PU (H11) and TTF (H12) positively impact users' continuous intention (CI) in most cases (except inexperienced users' PU). These may be trivial hypotheses to prove, but given some previous concerns regarding data quality collected from MTurk, the result of these hypotheses, including the path coefficient in both user groups, may serve as a control test ensuring our data quality.

Theoretical implications

Our primary theoretical implication is the critical success factors (CSFs) of customers' continuous intention of crypto-exchange adoptions (see Table 4, which includes a summary of our findings and possible explanations we made). Combining the characteristics of asset-light highlighted by previous FinTech literature, we accordingly proposed seven factors, namely the LAS-VICT principle ("L" stands for "low user-burden", "A" stands for "asset-light", "S" stands for

"scalability", "V" stands for "variety", "I" stands for innovativeness, "C" stands for compliance-sensitive and "T" stands for transparency.)

The validated characteristics were comparable to previous FinTech literature. For example, transparency is highly valued by users for both crypto-based and conventional FinTech (Jünger & Mietzner, 2020). In addition, we identified that users value the scalability of crypto-exchanges, which echoes previous literature related to conventional FinTech (e.g., Lee & Teo, 2015; Muthukannan et al., 2021).

Third, innovativeness is vital for both crypto-based and fiat-based FinTech (as indicated by LASIC principles). While according to our results, the direct influence of INN is insignificant on TTF (and marginally significant on PU), INN significantly influences all three kinds of service varieties, which, in turn, influences TTF and PU.

Moreover, while our study did not cover the asset-light aspect, we argue that asset-light is also a characteristic of successful crypto-exchanges, derived from previous literature. For example, similar to crypto-exchanges, customers of neobank and other FinTech are largely served via the Internet instead of firms' physical branches or infrastructure (Lee & Teo, 2015). Also, the study of Nurbaev et al. (2022) on neobank suggested that Coinbase (one of the largest

Table 4 Discussion on results related to tested characteristics of crypto-exchanges

Results	Explanations
Low user-burden	
The impact of financial costs is insignificant on both Perceived usefulness (PU) and Task-technology fit (TTF)	While prior research suggests that a low margin is essential for the success of FinTech (Arslanian & Fischer, 2019; Singh, 2020), we noted that this is not the case for crypto-exchanges. We suggest that low margin has become a by-default, taken-for-granted characteristic of online services in the era of platformization (Au, 2020). Costs play an important role as they are expected to be negligible for the customer. Demanding low costs is a necessity but will not provide any advantage over other exchange platforms, which is equivalent to the “must-be” quality of the Kano Model (Kano et al., 1984). Users expect low financial costs and would not be delighted if it happened, but they would feel unsatisfactory if it were not available
Financial benefits positively impact PU and TTF for the overall sample	Prior research suggests that providing financial benefits can encourage users to use online services, such as online payment services (See-To & Ho, 2016). For the case of crypto-exchanges, we noted that users welcome crypto-exchanges that benefit them financially and thus mediate financial costs
Scalability	
Impacts of perceived scalability are significant only on TTF for overall and experienced users	Prior research mentioned the positive impacts of scalability on developing FinTech ecosystems, which can benefit both PU and TTF (Muthukannan et al., 2021). Yet, in our findings, we only found that it happened only for TTF in crypto-exchanges. We noted that experienced users may encounter more issues due to the lack of scalability, such as temporary wallet inaccessibility, and thus more sensitive to scalability issues
Variety	
Perceived variety of deposit options (PVD) negatively and marginally influences perceived ease of use (PU) for inexperienced users, but its impact on PU is insignificant for experienced users or the entire group	It may be explained by the nature of crypto-exchanges as a financial service, which requires users to learn carefully before a more comprehensive adoption. For inexperienced users, offering too many deposit options may create confusion. Experienced users may be more informed and less confused, but they may, on the other hand, feel indifferent as long as there is an effective way to deposit the money
Perceived variety of wealth growth options (PVG) positively and significantly impact TTF. The impact is, however, insignificant when considering only experienced users	Less experienced users may find that a larger range of options can provide them with more different possible ways to earn However, experienced users are get used to different wealth growth options and may easily understand a new one upon learning about it. They may find that the outcomes of available wealth growth options are more important than the range of options offered
Perceived variety of withdrawal options (PVW) positively and significantly impacts TTF. PVW positively impacts but marginally PU for inexperienced users	Experienced users may consider cryptocurrencies a longer-term investment and have a less immediate need to withdraw the money. Even if they need to, they are already informed about how to do so. As long as there is one effective way, they are satisfied. Inexperienced users have greater concerns about withdrawal options as they are unsure whether they will adopt cryptocurrencies in the longer term. Thus, they may be more welcoming to a wide range of withdrawal options. However, both groups feel withdrawal options are vital for their task need

crypto-exchanges) had a profit-asset ratio of 1.15%, significantly lower than that of HSBC (one of the largest universal banks and financial service groups) at 5.1%.

However, we have also identified some characteristics of successful crypto-exchanges that are distinctive from other FinTech. First, financial benefits may be essential for crypto-exchange more than for other FinTech, such as m-payment, InsurTech, and lending services, despite possible inclusions of financial rebates or other perks in these more conventional FinTech. This is because the currencies involved in

the crypto-exchange transactions (i.e., cryptocurrencies) are usually, if not always, volatile in price. Holding cryptocurrencies may create risks from many users' perspectives, while temporal price drops may compromise profits from cryptocurrency investments. Conversely, these benefits may offset the potential loss or risk associated with cryptocurrency assets and investments. Users' emphasis on financial benefits may be reflected by path coefficients in our results (0.120 for H2b: FIN-B $\rightarrow$ TTF and 0.123 for H2a: FIN-B $\rightarrow$ PU, with $p < 0.05$ for both paths).

Table 4 (continued)

Results	Explanations
Innovativeness	
Perceived innovativeness significantly and negatively impacts PU among inexperienced users, but its impact is only marginal when viewing the entire sample	Prior research suggests that innovativeness has a positive impact on FinTech products (Gimpel et al., 2018; Lee & Teo, 2015; Liu et al., 2020). Yet, we observed a negative effect, probably due to the nature of crypto-exchanges. Too innovative options may create short-term confusion for inexperienced users. On the other hand, experienced users may have gotten used to these innovative options and have greater concerns about the actual outcomes of using the crypto-exchanges However, actual firms' innovativeness may drive service options variety development, which will positively impact TTF/PU. An ad hoc linear regression suggested that innovativeness and scalability significantly impact service varieties across different groups (see Table 5 for further information)
Compliance-sensitive	
Perceived legal compliance is insignificant on both TTF and PU	Regulators, but not users, may be concerned about firms' compliance. Reviewing the demographic information of respondents, we find that most come from countries with more relaxed regulations related to cryptocurrencies (e.g., USA and India). Thus, legal compliance issues may not become their concern
Perceived social compliance positively impacts PU in general. However, its impact is insignificant when viewing experienced users	While researchers argue that social compliance of FinTech companies will have positive impacts on the industry (Li et al., 2019), we only observed its influence partially. Some users who uphold ethical consumerism may find social-compliance products/services with higher PU. Its insignificant impact on experienced users may be because they are already well informed, have carefully considered any social issues related to the crypto-exchange, and thus have fewer concerns
Transparency	
Transparency positively impacted the TTF and PU of the crypto-exchanges for all cases, except for experienced users' PU	Transparency has been known as a key positive factor influencing the success of FinTech (Jünger & Mietzner, 2020; Nurbaev et al., 2022). In this study, we noted that transparentising operations and market trends may create a higher perceived value and mediate the issues of cryptocurrency knowledge and price volatility

Table 5 Ad hoc linear regression results for testing the impact of perceived innovativeness and scalability on perceived varieties of deposit, wealth growth, and withdrawal options

DV	IV	Overall users		Inexperienced users		Experienced users	
		Path coeff	p-value	Path coeff	p-value	Path coeff	p-value
PVD		Adjusted <i>R</i> -square: 50.0%		Adjusted <i>R</i> -square: 61.9%		Adjusted <i>R</i> -square: 32.9%	
	SCAL	0.602	< 0.001***	0.690	< 0.001***	0.478	< 0.001***
	INN	0.243	< 0.001***	0.257	< 0.001***	0.226	0.011*
PVG		Adjusted <i>R</i> -square: 42.6%		Adjusted <i>R</i> -square: 50.2%		Adjusted <i>R</i> -square: 29.2%	
	SCAL	0.523	< 0.001***	0.616	< 0.001***	0.378	< 0.001***
	INN	0.278	< 0.001***	0.245	< 0.001***	0.309	< 0.001***
PVW		Adjusted <i>R</i> -square: 32.9%		Adjusted <i>R</i> -square: 44.9%		Adjusted <i>R</i> -square: 17.2%	
	SCAL	0.384	< 0.001***	0.520	< 0.001***	0.199	0.044*
	INN	0.339	< 0.001***	0.328	< 0.001***	0.334	< 0.001***

1. DV, dependent variable; IV, independent variable; SCAL, perceived scalability; PVD, perceived variety of deposit options; PVG, perceived variety of wealth growth options; PVW, perceived variety of withdrawal options; INN, perceived Innovativeness

2. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$, # $p < 0.10$

Second, social compliance is another aspect relevant to crypto-exchanges but not necessarily other FinTech. It may be because cryptocurrencies are more often related to social issues, such as electricity wastage (Li et al., 2019), scams (Xia et al., 2020), and facilitating illegal activities (Brenig et al., 2015). Before adoptions, users may consider how crypto-exchanges respond to these issues. Such social compliance consideration may be reflected by, for example, the path coefficients of COMP-S $\rightarrow$ PU (H9a, $\beta=0.229$, with $p=0.005$). Conversely, when users consider their potential adoption of other FinTech applications, these social issues may not come into their consideration, especially when the subject FinTech applications are launched by some firms that they have already established trust (Pentina et al., 2013).

Third, our results suggested that perceived service variety can significantly drive continuous intention of adoption of crypto-exchanges, as indicated by the path coefficient of PVG $\rightarrow$ TTF (H4b) and PVW $\rightarrow$ TTF (H5b), both at significant levels with $p < 0.05$. While this echoes some previous FinTech literature (Leong et al., 2017; Muthukannan et al., 2021), we argue that varieties can be more important in the context of crypto-exchanges, given cryptocurrencies are more technically complicated than other FinTech (Fröhlich et al., 2021). Conversely, in some other FinTech, services offered by successful firms can be highly monopolistic (e.g., P2P lending platforms, see Au et al., 2020) with very limited variety. Along with customer education, varieties can simplify the complications and may enhance users' adoption (Au et al., 2022; Shieh et al., 2023).

Last but not least, to identify whether the majority of exchanges provide a comparable experience to the participants, we conducted an ad hoc comparison between respondents based on two leading crypto-exchanges (Binance and Coinbase, most selected by our respondents). From the entire dataset, we extracted the respondents who chose Binance and Coinbase to generate two data subsets. Each of the data subsets was used to test our theoretical model again. We compared the results and identified no significant difference between the testing results for most hypotheses (except H2b and H5a). For H2b, the path coefficient difference (Binance – Coinbase) is 0.340 with a 2-tailed p -value of 0.005, indicating a stronger role of financial benefits on task-technology fit in the context of Binance, compared to Coinbase. Nevertheless, H2b is supported for both data subsets. For H5a, the path coefficient difference (Binance – Coinbase) is 0.452 with a 2-tailed p -value of 0.016, indicating a stronger role of perceived variety of withdrawal on perceived usefulness in the context of Binance, compared to Coinbase. This may be explained by the far-higher variety of withdrawal options of Binance, which included P2P cryptocurrency trading and a higher range of supporting cryptocurrencies, in addition to the cryptocurrency debit cards (offered by both Binance and Coinbase).

Yet, solely using data subsets that include respondents from only one crypto-exchange is inadequate for fully validating our theoretical model, given the insufficient sample size when considering only one crypto-exchange (77 for Coinbase, 57 for Binance). This analysis allowed us to assume that two crypto-exchanges offer experiences comparable to those of the respondents in the meantime, but further testing with more respondents, especially on comparing different crypto-exchanges, is encouraged.

Practical implications

Our study provided some practical implications for crypto-exchanges. Based on the several identified factors that drive the development of crypto-exchanges, we suggest that, first, crypto-exchanges should develop a wider range of service options to cater to different customers' needs and thus enhance overall perceived usefulness. The development of these options may be enabled by firms' innovativeness, which may be enhanced by, for example, engaging external developers via API infrastructure and hackathons (Kitsios & Kamariotou, 2023) or other tactics. Given the increasing willingness of cryptocurrency adoptions among e-commerce merchants (Dyosi & Tanner, 2023), crypto-exchanges may collaborate with more innovative firms to enhance their service variety. However, confusion on service options can be an issue, which may require additional customer education effort.

Second, crypto-exchanges should be sensitive not only to legal requirements but also to social expectations. For example, they may explain to the public how cryptocurrencies may address social needs. They may also illustrate how the measures they adopt (e.g., KYC⁵ against money laundering) or their service options (e.g., cryptocurrency donations to charity, allowing quicker and less costly transactions) have mediated the dark side of cryptocurrencies and fostered the advantages of cryptocurrencies.

Third, ensuring transparency can provide customer confidence in these firms, which has become important when the cryptocurrency world has been shocked by various incidents, such as the fall of FTX and the collapse of Terra-LUNA. These incidents created panic in the market, especially when there was a lack of information, and some media even suggested the arrival of "Crypto Winter" (Kharpal & Browne, 2022; Powell, 2023). In turn, less capital is invested in cryptocurrencies, which means

⁵ Know-Your-Customer (KYC) refers to the processes of identifying customers for preventing those who wish to use the financial services systems for crime access, and has been recognized as an important aspect of fraud prevention (Gill and Taylor, 2004).

crypto-exchanges are more likely to have a difficult time. Conversely, crypto-exchanges that can assure their users about service stability and asset safety, possibly by maintaining information transparency, are more likely to survive in the winter.

Conclusion

Assuming crypto-exchanges as a distinctive type of FinTech, we grounded on previous FinTech literature and researched the characteristics of crypto-exchange business successes. We first proposed some characteristics that can drive crypto-exchanges' successes. These characteristics are, in turn, tested from the users' perspective. Based on the results, we have provided both theoretical implications, such as the difference between crypto-exchanges and other types of FinTech, and practical implications for crypto-exchange owners.

Future research may, first, include a large sample size with a more diversified respondent background to validate our implications. In particular, a larger sample size can allow us to explore further the potential differences between the major crypto-exchanges, which we cannot fully explore at the moment as some of the crypto-exchanges only have a handful of users participating in our survey. Second, future research may address the phenomenon from different perspectives beyond users' perceptions and service adoptions (possibly including more different types of data, such as actual users' trading data), as well as including other established variables (e.g., trust) from technology acceptance literature in the theoretical model. Moreover, we believe that if firms can realise the identified CSFs, users can recognise firms' efforts. However, we agree that perception-reality gaps may exist, and further studies are encouraged to explore further. For example, our measurements of crypto-exchanges' innovativeness and scalability were made from the end users' perspective, but other methods, such as third-party evaluation, may also measure these aspects. Researchers may also use indirect methods (e.g., Ho et al., 2020) to measure the "price tag" of some of the services and use it as a reference to estimate the incentives to provide to potential users to attract them to use such services. Furthermore, it will be desirable if future research can consider respondents' affiliations with crypto-exchanges, regardless of the methodologies adopted.

In the end, it is hoped that our study can contribute to crypto-exchanges' successes and the advancement of FinTech theories and provide valuable references for the industry.

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